PORTFOLIO WALKTHROUGH

CUSTOMER SHOPPING BEHAVIOR

From raw data to business insight.



Made with GAMMA

THE WORKFLOW

This project connects Python, SQL, and Power BI into one analytics workflow

Portfolio walkthrough of an end-to-end customer analytics project.

Goal: turn customer-level shopping data into findings for marketing, merchandising, and sales decisions.

01

Python + Pandas

Prepare and shape customer-level shopping data.

02

PostgreSQL + SQL

Turn attributes into business questions.

03

Power BI

Present findings as an interactive portfolio deliverable.

Tools: Python, Pandas, PostgreSQL, SQL, Power BI, and Jupyter Notebook.

Made with GAMMA

DATASET

3,900

customer records

18 original attributes provide a broad view of shopping behavior.

3,900 customer records create a broad view of shopping behavior



Customer mix

Demographics, subscriptions, and purchase frequency.



Purchase behavior

Products, purchase amounts, discounts, shipping, and payment.



Experience signals

Ratings and customer-level differences.

Business questions: who drives revenue, what products perform, how promotions behave, and where customer experience differs.

Made with GAMMA

Data preparation turned a messy source into an analysis-ready table

01

Inspect

Reviewed dimensions, types, descriptive statistics, and missing values.

02

Impute

Filled 37 missing Review Rating values with the median rating within each product category.

03

Standardize

Converted columns to lowercase snake_case and renamed purchase_amount.

04

Deduplicate

Compared discount_applied and promo_code_used, found identical information, and removed the redundant field.

Made with GAMMA

FEATURE ENGINEERING

Feature engineering made customer behavior easier to compare

Raw attributes became comparable business features.

Python/Pandas

Engineered Features

PostgreSQL/SQL

- Created age groups to support demographic segmentation
- Mapped purchase-frequency labels such as Weekly, Fortnightly, Quarterly, and Annually into approximate day intervals
- Loaded the prepared data into PostgreSQL through SQLAlchemy for repeatable business analysis



Made with GAMMA

SQL converted raw attributes into business questions



Revenue mix

Revenue by gender and age group.



Promotion behavior

Discount usage and discounted-purchase concentration.



Product demand

Top-rated and most frequently purchased products.



Customer patterns

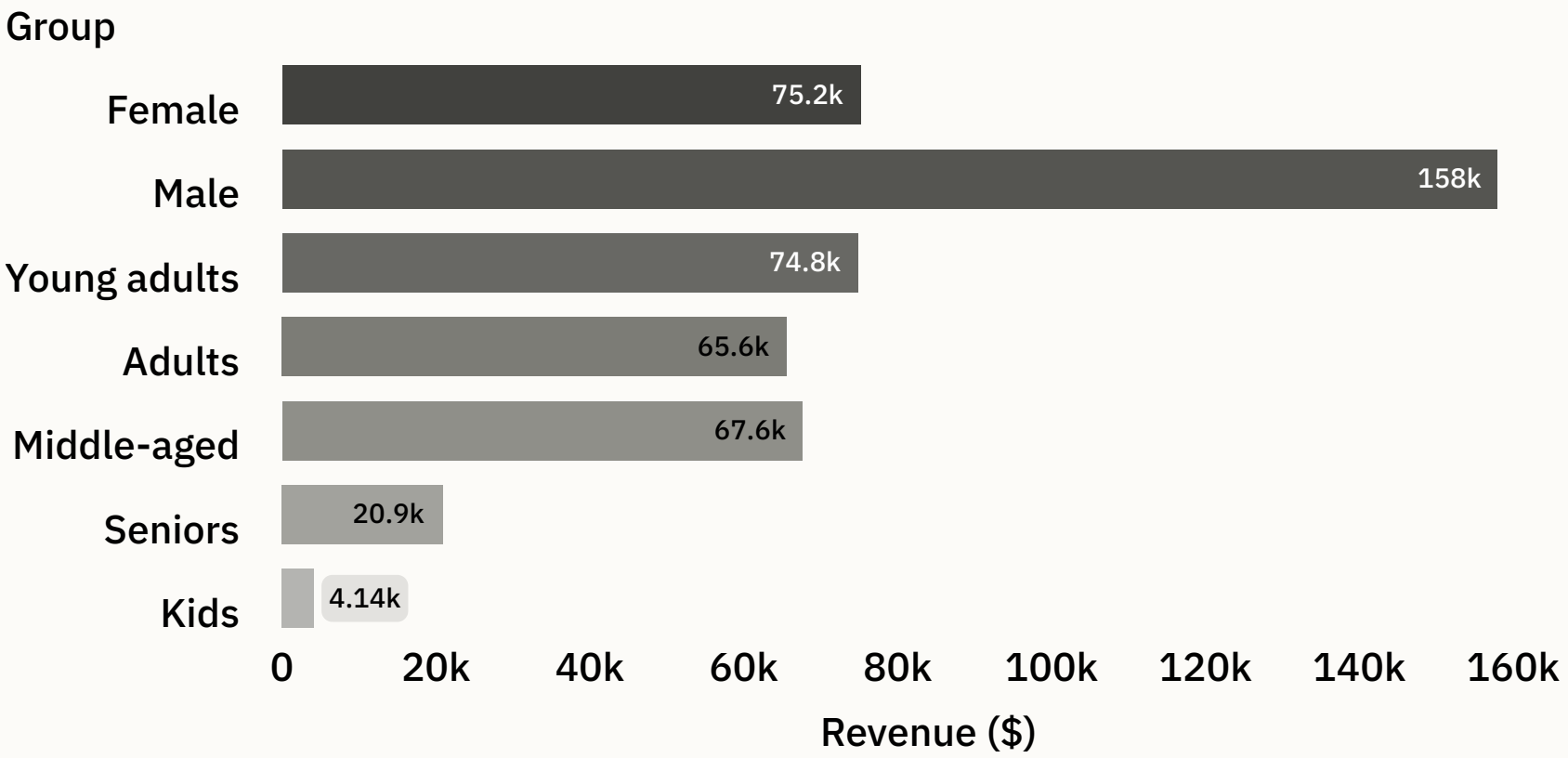
Shipping-method, subscription, and previous-purchase comparisons.

Practical SQL fluency: aggregations, conditional logic, subqueries, CTEs, and window functions.

Made with GAMMA

Revenue concentration by customer group

The strongest revenue signal comes from customer mix, not just product choice



Observed in this dataset

$233,081

Recorded revenue totaled $75,191 for female customers and $157,890 for male customers.

Age-group contribution: young adults 32%, middle-aged customers 29%, adults 28%, seniors 9%, and kids 2%.

Descriptive results, not causal explanations.

Made with GAMMA

Product rankings and repeat-buyer segments reveal where to focus next

Top-rated products	Most frequent purchases by category	Customer segments
Gloves 3.86	Jewelry	**3,116** Loyal
Sandals 3.84	Blouse	**701** Returning
Boots 3.82	Pants	**83** New customers
Hat 3.80	Sandals	
Skirt 3.78	Shoes	
	Sneakers	
	Jacket	
	Coat	

Ratings show what customers value. Frequency shows where repeat demand is already visible

Made with GAMMA

PORTFOLIO DELIVERABLE

Power BI turns the analysis into an interactive portfolio deliverable

A viewer can move from a high-level KPI to a customer group, product category, or purchasing pattern.

No completed dashboard image was included. The deliverable is shown as a clean wireframe and native analytical components.

Dashboard logic

KPI

$233K

Recorded revenue

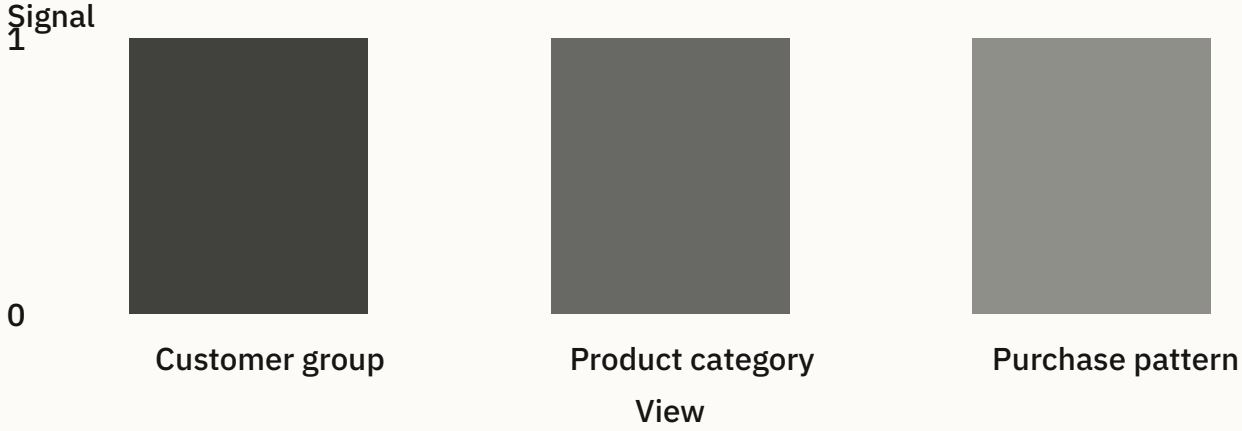
SEGMENTS

3

Loyal · Returning · New

SLICERS

Group · Product · Pattern



Categorical comparisons, bar and column charts, a donut chart, and interactive slicers.

Made with GAMMA

FROM ANALYSIS TO ACTION

The project demonstrates a repeatable path from data cleaning to commercial recommendations

01 ## Segment marketing

Use purchasing history and customer mix to shape marketing priorities.

02 ## Guide merchandising

Use product ratings and purchase frequency to inform merchandising priorities.

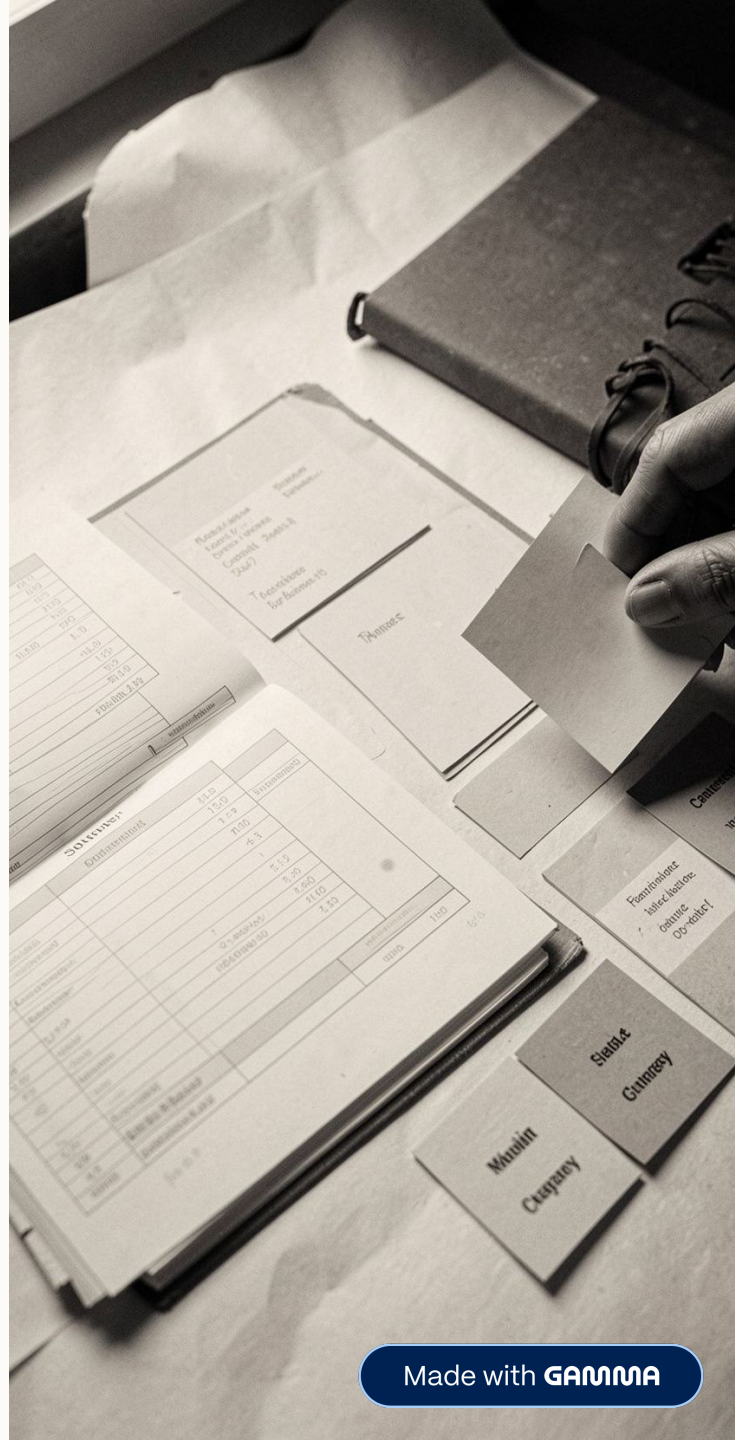
03 ## Test promotion signals

Evaluate discount behavior alongside purchase amounts before calling promotions effective.

04 ## Extend the analysis

Investigate shipping and subscription differences without treating correlation as causation.

Next analytical step: add dated transactions, margin data, and cohort tracking to move from descriptive analysis toward retention and profitability.



Made with GAMMA